AN INTERNSHIP REPORT

Of

1-MONTH JAVA DEVELOPER INTERNSHIP

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering with Specialization in Artificial Intelligence and Machine Learning)

By

Roll Number of Student: 25SCS1003004010

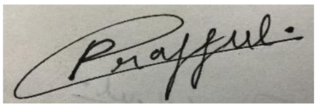
Name of Student: PRAFFUL SHIKHAR

Batch: 2025-2029

CANDIDATE'S DECLARATION

I, **PRAFFUL SHIKHAR**, do hereby declare that the internship report titled **"1-Month Java Developer Internship"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering with Specialization in Artificial Intelligence and Machine Learning.

All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: __________________________

Student Name: PRAFFUL SHIKHAR
Roll No.: 25SCS1003004010

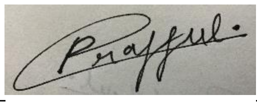
ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their inspiring guidance, invaluably constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to express my sincere appreciation and gratitude to the entire team and technical mentors at **CodecTechnologies** for providing me with the opportunity and the structured industrial platform to learn and gain comprehensive hands-on exposure in the domain of enterprise Java development, Spring Boot, Hibernate ORM, and database engineering.

I am also profoundly thankful to my institute, **IILM University, Greater Noida**, and the esteemed faculty members of the **School of Computer Science and Engineering** for their continuous support, academic guidance, and motivation throughout the duration of this program.

I express my deepest gratitude to my parents for their blessings, encouragement, and unconditional support.

Signature: ___________________________

Student Name: PRAFFUL SHIKHAR
Roll No.: 25SCS1003004010

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3. INTERNSHIP COMPLETION CERTIFICATE & OFFER LETTER

INTERNSHIP COMPLETION CERTIFICATE

CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

MINISTRY OF EDUCATION

CERTIFICATE

OF INTERNSHIP

This certificate is awarded to

Prafful Shikhar

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC Approved internship program as

Java Developer Intern

Conducted from 01/06/2026 to 01/07/2026

Google for Education Partner

AICTE ID - CORPORATE6759d549ce59e1733940553

Dr. Anurag Shrivastava

Program Manager

INTERNSHIP OFFER LETTER

CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

INTERNSHIP OFFER LETTER

Dear Prafful Shikhar,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Java Developer Intern
- Location : Hybrid / India
- Duration : 01/06/2026 to 01/07/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by Project Heads.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

Best regards,

Sincerely,
Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 01/06/2026

Dr. Anurag Shrivastava
Talent Acquisition Manager

Google for Education Partner

ISO 9001 CERTIFIED

www.codectechnologies.in support@codectechnologies.in Chandivali IT hub, Mumbai

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed technical account of the 1-Month Java Developer Internship undertaken at **CodecTechnologies**. The internship was carried out from **01/06/2026 to 01/07/2026** as part of the academic requirements for the Bachelor of Technology in Computer Science and Engineering with Specialization in Artificial Intelligence and Machine Learning at **IILM University, Greater Noida**.

The primary aim of this internship was to build a robust, industry-aligned foundation in enterprise backend development, object-oriented software engineering, and modern Java frameworks. The curriculum bridged core programming methodologies with enterprise architectural paradigms, focusing on the Spring ecosystem (Spring Boot, Spring MVC, Spring Data JPA), Object-Relational Mapping (Hibernate), relational database design (MySQL), and dynamic template rendering (Thymeleaf, HTML5, CSS3).

Over the course of the internship, two comprehensive, production-grade applications were designed, implemented, and tested:

- **Employee Management System:** An enterprise web application designed to handle organizational workforce records, departmental mappings, role-based access control (RBAC), and transactional CRUD operations using Spring MVC, Hibernate ORM, and MySQL.
- **Student Management System:** An academic information portal built to track student profiles, course enrollments, examination marks entry, automated academic performance metrics, and report generation using Spring Boot, Spring Data JPA, MySQL, and Thymeleaf.

4.2 Organization Profile

Codec Technologies is an established software engineering solutions provider and technical talent development firm specializing in delivering scalable digital solutions, corporate training, and experiential internship programs for engineering undergraduates. Codec Technologies delivers structured development curricula designed to mirror corporate Agile software development lifecycles (SDLC). The organization emphasizes industry-standard design patterns, dependency injection mechanisms, modern persistence layers, code refactoring best practices, and collaborative version control workflows.

4.3 Problem Statement

In modern corporate environments and educational institutions, managing large volumes of entity records manually or via legacy disconnected spreadsheets leads to substantial data redundancy, synchronization inconsistencies, lack of access control, and human errors. Furthermore, engineering students frequently learn foundational Java concepts in isolation (such as syntax and basic algorithms) without adequate exposure to layered enterprise architecture, automated persistence engines, transactional integrity, and Model-View-Controller design patterns.

The problem addressed during this internship was to design and implement secure, decoupled, and database-driven enterprise web applications that streamline administrative operations, enforce rolebased access governance, eliminate data anomalies through ORM validation, and provide seamless, responsive user interfaces.

4.4 Project Objectives

The core technical objectives accomplished during the 1-Month Java Developer Internship include:

- To master the fundamentals of enterprise application development using Core Java (JDK 17/21), Object-Oriented Programming (OOP) paradigms, and the Collections Framework.
- To design and implement multi-tiered applications using the **Model-View-Controller (MVC)** architectural pattern via **Spring Boot** and **Spring MVC**.
- To architect optimized relational database schemas in **MySQL** enforcing entity constraints, foreign keys, and indexing.
- To implement automated Object-Relational Mapping (ORM) and persistent data layers using **Hibernate** and **Spring Data JPA**.
- To construct complete **CRUD (Create, Read, Update, Delete)** pipelines with transactional safety for administrative record management.
- To enforce **Role-Based Access Control (RBAC)** and server-side data validation to secure confidential records.
- To design dynamic, responsive front-end user interfaces utilizing **Thymeleaf** template engine, **HTML5**, and **CSS3**.
- To manage project builds, modular plugins, and external dependencies using **Apache Maven**.
- To maintain rigorous version control, repository branching, and issue tracking using **Git** and **GitHub**.

4.5 Scope of the Project

The scope of the internship encompasses the entire backend and frontend engineering lifecycle of two enterprise management systems:

- **Employee Management System:** Scoped around managing enterprise workforce data, covering employee onboarding, departmental allocation, role classification (Admin/Manager/Employee), salary bracket updates, profile search/filtering, and record lifecycle termination.
- **Student Management System:** Scoped around academic institutional administration, handling student profile enrollment, course mappings, semester-wise marks input, GPA/percentage autocalculation, and dynamic printable report generation.

The project execution was conducted within local server development environments, utilizing embedded Tomcat servlet containers, MySQL database servers, and Maven build pipelines, strictly adhering to clean code standards and scalable software engineering practices.

4.6 Technologies and Tools Used

- **Core Programming Language:** Java (JDK 17 / JDK 21)
- **Backend Frameworks:** Spring Boot, Spring MVC, Spring Data JPA

• **ORM & Persistence:** Hibernate ORM, JDBC API

• **Database Management System:** MySQL Server 8.0, MySQL Workbench

• **Template Engine & Frontend:** Thymeleaf, HTML5, CSS3, JavaScript

• **Build & Dependency Management:** Apache Maven

• **Version Control & Collaboration:** Git, GitHub

• **Integrated Development Environments (IDEs):** IntelliJ IDEA, Eclipse, Visual Studio Code

4.7 System Architecture

The software systems developed during the internship utilize a decoupled, multi-tiered **Layered Architecture** adhering to the Model-View-Controller (MVC) pattern. This ensures maintainability, high cohesion, low coupling, and scalability across components.

```
+-----------------------------------------------------------------------+ |
PRESENTATION / VIEW LAYER | | (Thymeleaf Templates, HTML5, CSS3 Responsive UI
Forms) |
+-----------------------------------------------------------------------+ │ ▼
[HTTP Requests / Form Submissions]
+-----------------------------------------------------------------------+ |
CONTROLLER / ROUTING LAYER | | (Spring MVC Controllers / Spring Boot Web
Endpoints) |
+-----------------------------------------------------------------------+ │ ▼
[Service Invocation & DTOs]
+-----------------------------------------------------------------------+ |
SERVICE / BUSINESS LAYER | | (Business Rules, Marks Computation, Validation, RBAC
Security) |
+-----------------------------------------------------------------------+ │ ▼
[Entity Mapping & Transactions]
+-----------------------------------------------------------------------+ |
DATA ACCESS / PERSISTENCE LAYER | | (Spring Data JPA Repositories / Hibernate
ORM) | +-----------------------------------------------------------------------
+ │ ▼ [SQL Execution via JDBC Connection]
+-----------------------------------------------------------------------+ |
DATABASE / STORAGE LAYER | | (MySQL Relational Database Tables & Keys) |
+-----------------------------------------------------------------------+
```

Detailed Layer Descriptions:

1. **Presentation Layer:** Provides an intuitive user interface created with HTML5 and CSS3. Dynamic data binding is achieved via Thymeleaf expressions (e.g., `th:each`, `th:field`, `th:if`), enabling seamless data rendering from backend models.
2. **Controller Layer:** Manages web request endpoints, processes incoming HTTP GET/POST/PUT/DELETE requests, binds form payloads to Java Bean objects, triggers business services, and routes the client to appropriate views.
3. **Service Layer:** Acts as the core business logic hub. Implements transactional boundaries (`@Transactional`), enforces role-based permissions, performs input sanitization, and executes calculations (e.g., student grade point aggregation).
4. **Data Access Layer:** Leverages Spring Data JPA interfaces extending `JpaRepository`, eliminating boilerplate SQL queries through automatic query generation, ORM entity state management, and Hibernate session caches.
5. **Database Layer:** Consists of a normalized MySQL relational schema containing structured tables, primary keys, indexing, and foreign key cascades to ensure ACID compliance.

4.8 Methodology

The internship followed a structured, agile-inspired week-wise development methodology combining conceptual sessions, code reviews, schema modeling, lab implementation, and testing milestones:

Week	Module / Focus Area	Technical Description & Tasks Completed
Week 1	**Core Java Refinement & Spring Core Foundations**	In-depth revision of OOP concepts (Polymorphism, Inheritance, Encapsulation, Abstraction). Advanced Java features including Collections Framework, Generics, Stream API, and Exception Handling. Setup of IDEs, Maven configuration, and MySQL. Introduction to Spring Framework, Inversion of Control (IoC), and Dependency Injection (DI).
Week 2	**Employee Management System (Spring MVC & Hibernate)**	Designing relational database schemas for organizational employees and departments. Configuring Hibernate `hibernate.cfg.xml` and entity mappings (`@Entity`, `@Table`, `@Id`, `@Column`). Developing Spring MVC Controllers and implementing complete CRUD operations with Role-Based Access Control (Admin/Manager roles).
Week 3	**Student Management System (Spring Boot & JPA)**	Bootstrapping Spring Boot application using Spring Initializr. Implementing Spring Data JPA repository interfaces for student records, enrollment numbers, and course mappings. Designing the business service layer for marks computation, GPA grading algorithms, and transactional updates.
Week 4	**Thymeleaf UI Integration, Testing & Project Submission**	Developing responsive web pages with HTML5, CSS3, and Thymeleaf server-side templates. Integrating form validation, alert notifications, and search filters. Performing unit tests, database integrity verification, code refactoring, Git version control commits, and final project documentation.

Final	**Internship Evaluation & Credential Validation**	Comprehensive project evaluation by CodecTechnologies mentors, code quality review, and final issuance of the Certificate of Internship Completion.

Note: *Regular weekly milestones and code reviews were submitted to ensure standard industry compliance and robust application architecture.*

4.9 Expected Outcomes

Upon successful completion of the 1-Month Java Developer Internship at CodecTechnologies, the following key outcomes and technical competencies were achieved:

- **Practical Mastery of Enterprise Java:** Acquired deep practical understanding of the modern Java software stack, including Spring Boot, Spring MVC, Hibernate ORM, and Spring Data JPA.
- **Full-Stack CRUD Implementation:** Successfully designed and deployed two full-stack applications (Employee Management System and Student Management System) capable of handling comprehensive create, read, update, and delete database operations.
- **Secure Data Governance:** Implemented Role-Based Access Control (RBAC) preventing unauthorized modifications of employee designations and salaries.
- **Automated Academic Performance Calculation:** Built computational logic inside the service layer to automate marks aggregation, GPA calculation, and dynamic student report card generation.
- **Database Schema & Performance Optimization:** Developed normalized MySQL database schemas with foreign key relationships, transactional rollback protection, and optimized query execution.
- **Modern Dynamic UI Integration:** Seamlessly integrated Thymeleaf server-side templating with HTML5 and CSS3, producing responsive and intuitive administrative dashboards.
- **Professional Tooling & Version Control:** Gained proficiency with Maven for automated build lifecycles and Git/GitHub for branching, commit hygiene, and collaborative code distribution.
- **Credential Fulfillment:** Successfully completed all internship assessment tasks and received the official Certificate of Internship Completion from CodecTechnologies.

4.10 Certificates of Completion and Communication Proof

The official Internship Offer Letter and Certificate of Completion issued by **CodecTechnologies**, confirming successful completion of the 1-Month Java Developer Internship from 01/06/2026 to 01/07/2026, are documented in Chapter 3.

Below are the additional proofs of communication, credential issuance notifications, and project repository records confirming active development:

COMMUNICATION PROOF :

Welcome to Codec Technologies...

Hi Prafful,

Just to let you know — we've received your order #226280, and it is now being processed:

[Order #226280] (May 11, 2026)

Product	Quantity	Price
1-Month Java Developer Internship	1	₹99.00

Subtotal:	₹99.00
Payment method:	UPI/Paytm/Credit Card/Debit Card/NetBanking
Total:	₹99.00

Billing address

Prafful Shikhar
17-B ,Flat No. A4 , Sewa Bhagya Narayan Enclave, Near Vishal Mega Mart ,

Submission of Completed Projects-Codec Technologies Internship

The Phoenix Gamer YT <blenderguy53@gmail.com>
to vaishali

Respected Ms. Vaishali,
I have completed the two projects which were given in my Java Developer Internship.
I would like to attach the link of my two GitHub Repositories for you to check:
1. Student Management System
GitHub Repository Link: https://github.com/CodesByPrafful/Student-Management-System

2. Employee Management System
GitHub Repository Link: https://github.com/CodesByPrafful/Employee-Management-System

I have also attached the Offer Letter in this email for your reference.
Thank you for the opportunity.I look forward to your feedback.

Regards,
Prafful Shikhar
Email:blenderguy53@gmail.com
Phone Number:7518923495

GITHUB REPOSITORY LINK OF THE PROJECTS:

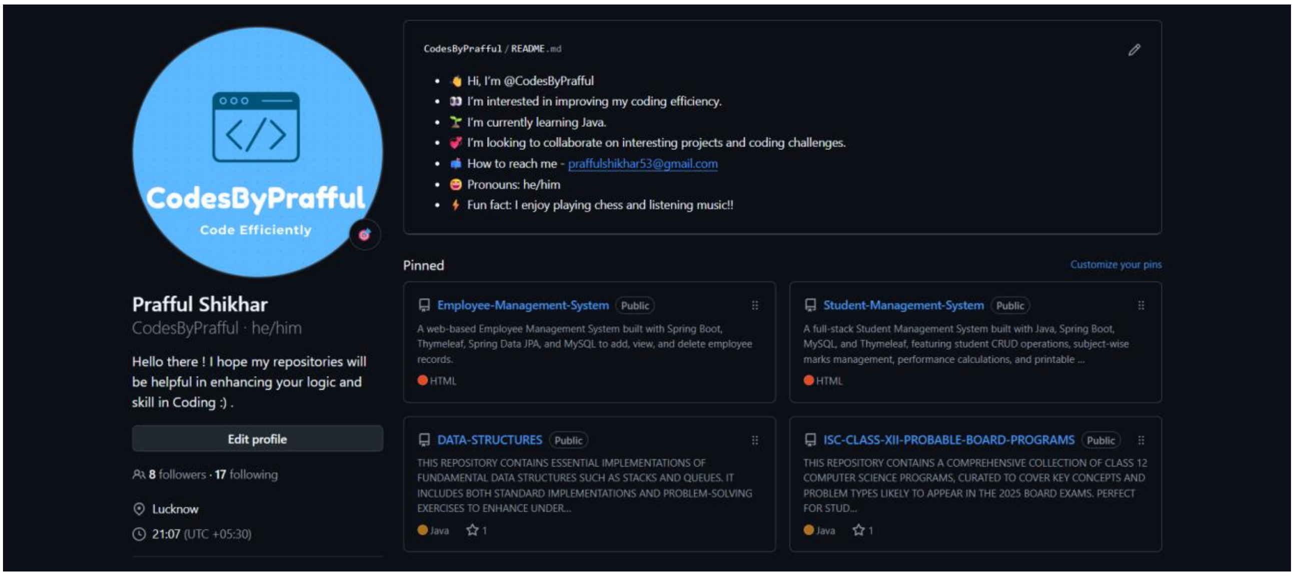
1. **STUDENT MANAGEMENT SYSTEM:** A full-stack Student Management System built with Java, Spring Boot, MySQL, and Thymeleaf, featuring student CRUD operations, subject-wise marks management, performance calculations, and printable student reports.

   LINK: https://github.com/CodesByPrafful/Student-Management-System

2. **EMPLOYEE MANAGEMENT SYSTEM:** A web-based Employee Management System built with Spring Boot, Thymeleaf, Spring Data JPA, and MySQL to add, view, and delete employee records.

   LINK: https://github.com/CodesByPrafful/Employee-Management-System

PROOF :



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